

Functional Annotation Report: *dacA* / PP_4803 (UniProt Q88DM2)

Organism: *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950), taxid 160488 **Gene:** *dacA* (ordered locus PP_4803) — EMBL AAN70372.1 **Protein:** serine-type D-Ala-D-Ala carboxypeptidase, EC 3.4.16.4 **Family:** Peptidase S11 (Pfam PF00768 + PF07943); penicilloyl-serine transferase / β -lactam-transpeptidase-like fold (InterPro IPR012338, IPR015956, IPR018044, IPR012907, IPR037167) **Length:** 386 aa (Sec signal peptide 1–24; mature chain 25–386)

1. Summary (answer to the research question)

dacA (PP_4803) encodes a **low-molecular-weight penicillin-binding protein (LMW-PBP)**, specifically a **serine-type D-alanyl-D-alanine carboxypeptidase (DD-carboxypeptidase; EC 3.4.16.4)** and the *P. putida* ortholog of *Escherichia coli* PBP5. Its primary reaction is **hydrolytic removal of the terminal D-alanine from the C-terminal D-Ala-D-Ala of muramyl-pentapeptide stems in peptidoglycan**, converting pentapeptides to tetrapeptides. By depleting the pentapeptide donor substrate used by transpeptidases, it **regulates (limits/tunes) the degree of peptidoglycan cross-linking**, which is required for normal cell shape and for buffering the envelope against β -lactam stress. The enzyme acts **in the periplasm on the peptidoglycan sacculus**, tethered to the outer face of the inner (cytoplasmic) membrane by a C-terminal amphipathic anchor after Sec-dependent export. It functions in the **peptidoglycan biosynthesis / cell-wall maturation pathway**, and — like its *Pseudomonas aeruginosa* ortholog — its penicilloyl-serine fold additionally supports intrinsic β -lactam resistance.

Note on evidence base: No experimental study has been published on PP_4803 itself. The functional assignment is high-confidence and rests on (i) unambiguous sequence/domain identity, (ii) conservation of all catalytic motifs, and (iii) an extensive body of

experimental work on the orthologous PBP5/DD-carboxypeptidase enzymes of *E. coli* and *P. aeruginosa*. Claims are labeled as direct database annotation, bioinformatic inference, or ortholog-based experimental evidence.

2. Identity verification

The gene symbol, EC number, organism, family, and domain architecture from UniProt are internally consistent and match the peptidase-S11 DD-carboxypeptidase (PBP5) group. This is **not** the ambiguous "dacA" of Firmicutes (the di-adenylate cyclase / c-di-AMP synthase, an unrelated protein); the peptidase-S11 fold, EC 3.4.16.4, and the "dacA/PBP5" nomenclature in Gram-negatives are distinct and correct here.

Bioinformatic confirmation (this work): the 386-aa sequence contains **all three canonical penicillin-recognizing active-site motifs** of the penicilloyl-serine transferase superfamily:

Motif	Residues in Q88DM2	Role
SxxK	Ser64–Leu–Thr–Lys67	Catalytic nucleophile (Ser64 forms acyl-enzyme) + general base (Lys67)
SxN	Ser124–Gly–Asn126	Active-site cleft / oxyanion & substrate positioning
KTG	Lys226–Thr–Gly	D-Ala-D-Ala substrate-binding (β 3 strand)

UniProt independently annotates active sites at Ser64 (acyl-ester intermediate), Lys67 (proton acceptor), position 124, and a binding site at position 226 — in exact agreement. An N-terminal Sec signal peptide (1–24) and a hydrophobic/amphipathic C-terminus (...FFRRMWDSIRLFFYGLFN) complete the classic PBP5 topology.

3. Key findings (with evidence)

3.1 Primary reaction and mechanism

- **Reaction:** DD-carboxypeptidase — preferential cleavage of (Ac)₂-L-Lys-D-Ala-|-D-Ala; "removes C-terminal D-alanyl residues from sugar-peptide cell wall precursors" (UniProt catalytic activity / FUNCTION, EC 3.4.16.4). In peptidoglycan terms it removes the terminal D-Ala from the pentapeptide stem (L-Ala-D-Glu-*meso*-DAP-D-Ala-**D-Ala**), yielding a tetrapeptide.
- **Mechanism:** a two-step acyl-enzyme (ping-pong) mechanism via the Ser64 nucleophile — the same chemistry that makes these enzymes covalent targets ("penicillin-binding proteins") of β -lactams.
- **Ortholog evidence:** *E. coli* PBP5 "cleaves the terminal D-alanine from the muramyl pentapeptide in the peptidoglycan layer... In doing so, it varies the substrates for transpeptidation and plays a key role in maintaining cell shape" (Skoog et al.,

P 21674665

In Gram-negatives, DD-CPases generate the majority tetrapeptide muropeptides (Tehrani et al.,

P 40777447

3.2 Substrate specificity

- Specificity is centered on the **terminal D-Ala-D-Ala** dipeptide: "Beyond the C-terminal d-alanyl-d-alanine moiety, PBP5... has little specificity for other elements of peptidoglycan structure" (Nemmara et al.,

P 27420403

- It belongs to the membrane-bound **LMMA class** of DD-peptidases, which "catalyze DD-carboxypeptidase and endopeptidase reactions in vivo" and — unlike the soluble R39/R61 enzymes — do not require a free stem-peptide N-terminus, consistent with acting on the polymeric sacculus in situ (Nemmara et al.,

P 22029692

3.3 Subcellular localization / site of action

- **Periplasm, anchored to the outer leaflet of the inner membrane.** The Sec signal peptide exports the protein; the C-terminal amphipathic helix anchors it so the catalytic domain projects into the periplasm where peptidoglycan resides (the "periplasm... contains the peptidoglycan layer";

P 40777447

- *E. coli* PBP5 was characterized "in its native environment, the inner membrane," where it forms homo-oligomers (likely dimers) that position the catalytic domain to reach pentapeptides at varying distances from the membrane (Skoog et al.,

P 21674665

P. aeruginosa PBP5 was solubilized "by deletion of its C-terminal membrane anchor" (Smith et al.,

P 23629710

confirming the same topology in *Pseudomonas*.

- **Direct sequence-based corroboration (this work):** Kyte–Doolittle hydropathy analysis of Q88DM2 shows a hydrophobic N-terminal Sec signal peptide (residues 1–24, mean KD $\approx +1.27$) and a C-terminal **amphipathic** segment (residues 367–386, ... GGFFRRMWDSIRLFFYGLFN) — a basic Arg-Arg doublet (positive-inside) followed by an aromatic/aliphatic run — i.e., a monotopic amphipathic membrane anchor rather than a fully hydrophobic transmembrane helix. This is the diagnostic PBP5 topology and independently rules out a cytoplasmic or polytopic-transporter localization.

3.4 Biological process / pathway

- Pathway: **cell-wall biogenesis; peptidoglycan biosynthesis / maturation** (UniProt PATHWAY). GO: serine-type D-Ala-D-Ala carboxypeptidase activity (GO:0009002); regulation of cell shape (GO:0008360); cell-wall organization (GO:0071555); peptidoglycan biosynthetic process (GO:0009252).
- Mechanistically, by trimming pentapeptides to tetrapeptides the enzyme **controls the pool of transpeptidation donor substrate**, thereby setting the density/pattern of DD-cross-links. This is a *regulatory/maturation* role, not a wall-building (synthetic) one.

3.5 Physiological role (precise, not pleiotropic)

- **Cell-shape maintenance and intrinsic β -lactam resistance.** In the *E. coli* ortholog, *dacA*/PBP5 deletion sensitizes cells 4–8-fold to all β -lactams tested, reversibly (Sarkar et al., [P 20047819](#)), and the DD-carboxypeptidase catalytic activity is *essential* for this intrinsic resistance (Sarkar et al., [P 21719544](#)).
- **Functional redundancy:** bacteria encode several DD-CPases that act redundantly for robust cell-shape maintenance across conditions (Peters et al., [P 27329754](#)); hence *dacA* is typically individually non-essential, with defects unmasked only when paralogs are also removed.

3.6 *Pseudomonas*-specific / evolutionary context

- The closest structurally characterized ortholog, *P. aeruginosa* PBP5, is "one of the most abundant PBPs"; "its main function is that of a cell wall dd-carboxypeptidase, [but] it possesses sufficient β -lactamase activity to contribute to... resist... β -lactams," and its 2.05-Å structure "shows features closely resembling those of the class A β -lactamases" (Smith et al., [P 23629710](#)). This documents (i) the DD-CPase/ β -lactamase dual activity likely shared by *P. putida* DacA and (ii) the evolutionary relationship between DD-carboxypeptidases and class A β -lactamases.
- Ecological relevance in *P. putida*: this soil bacterium actively edits its peptidoglycan with non-canonical D-amino acids (e.g., via its broad-spectrum racemase acting on plant-derived amino acids such as canavanine; Aliashkevich et al., [P 33830599](#)). LMW-PBPs including PBP4/PBP5 can process peptidoglycan bearing non-canonical D-amino acids (Miyamoto et al., [P 32108264](#)), placing DacA within *P. putida*'s cell-wall chemical-editing physiology.

4. Supported and refuted hypotheses

Supported - H1: Q88DM2 is a serine DD-carboxypeptidase of peptidase family S11 (PBP5 ortholog) — supported by domain architecture and full catalytic-motif conservation. - H2: Its reaction is removal of the terminal D-Ala from muramyl-pentapeptide, regulating cross-linking — supported by UniProt catalytic activity + ortholog data. - H3: It localizes to the periplasm/inner-membrane and acts on the sacculus — supported by signal peptide, C-terminal amphipathic anchor, and ortholog membrane studies. - H4: It contributes to cell shape and intrinsic β -lactam resistance — supported by ortholog knockouts.

Refuted / rejected alternatives - Not the Firmicute *dacA* di-adenylate cyclase (c-di-AMP synthase): different fold, EC, and function — ruled out by family/domain identity. - Not a wall-synthesizing transpeptidase (HMW-PBP): it is a LMW-PBP carboxypeptidase that *limits* cross-linking substrate, not a primary polymerizing/cross-linking enzyme. - Not a soluble cytoplasmic enzyme: signal peptide + C-terminal membrane anchor place it in the periplasm.

5. Limitations and future directions

- **No direct experimental data on PP_4803**; conclusions are inferred from strong orthologs (*E. coli* PBP5, *P. aeruginosa* PBP5) and bioinformatics. Direct biochemical kinetics (kcat/Km on pentapeptide and β -lactams), a crystal/cryo-EM or AlphaFold-validated structure, and a *P. putida* $\Delta dacA$ (single and multi-CPase) phenotype would confirm specificity, dual β -lactamase activity, and the precise contribution to morphology.
- The **quantitative relative contribution of DacA vs. other *P. putida* DD-CPases** (redundancy structure) is unknown.
- Whether DacA preferentially processes **non-canonical D-amino-acid-substituted peptidoglycan** in *P. putida*'s soil/rhizosphere niche is an open, testable question.

References (PMIDs)

- 21674665 — Skoog et al. PBP5 homo-oligomer in the *E. coli* inner membrane; reaction and localization.

- 40777447 — Tehrani et al. Gram-negative DD-CPases generate tetrapeptides; periplasmic PG.
- 27420403 — Nemmara et al. PBP5 substrate specificity centered on terminal D-Ala-D-Ala.
- 22029692 — Nemmara et al. Substrate specificity/classification of LMM DD-peptidases.
- 20047819 — Sarkar et al. *dacA*/PBP5 deletion → β -lactam sensitization.
- 21719544 — Sarkar et al. PBP5 DD-CPase activity essential for intrinsic β -lactam resistance.
- 27329754 — Peters et al. Redundancy of PG carboxypeptidases for cell-shape robustness.
- 23629710 — Smith et al. *P. aeruginosa* PBP5 structure; DD-CPase + β -lactamase; membrane anchor.
- 33830599 — Aliashkevich et al. *P. putida* PG chemical editing with non-canonical D-amino acids.
- 32108264 — Miyamoto et al. LMW-PBPs process PG containing non-canonical D-amino acids.

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